

PramaanCV

Offline checks before trusting a vision model

Problem Statement ID

SIH26228

Problem Statement Title

Trustworthy Computer Vision Integrity Assurance for Data, Models and Inference Outputs in Multi-Contributor Pipelines

Theme: Blockchain & Cybersecurity

PS Category: Software

Team Name: PPLINTH Team ID: Not applicable





IDEA TITLE

PramaanCV: an offline assurance workflow

Proposed solution

Inspect contributed data and model evidence.
Protect results with a signed record.

How it addresses the problem

Show each finding and its affected asset.
Recommend acceptance, review or quarantine.

Innovation and uniqueness

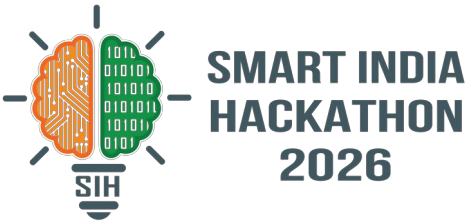
Proposed: a data finding guides a model test.
Linked evidence supports the analyst's decision.



Concept illustration



TECHNICAL APPROACH



Technologies to be used

Built: HTML/CSS, JavaScript, Node.js, SHA-256, Ed25519, JSONL
Next: isolated Python/PyTorch worker, ONNX adapter, durable storage

Methodology and process for implementation

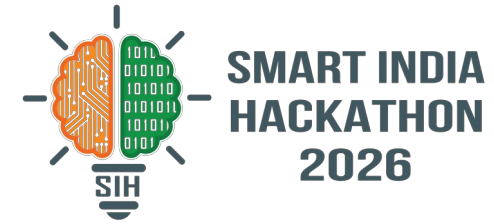
01	02	03	04
DATA	MODEL	RECEIPT	EVIDENCE
COCO label checks Source-level counts	Supplied output deltas Declared thresholds	File fingerprints Digital signature	Reasons and findings Rule-based decision

Working prototype: sign a record, change its output digest, verify failure

Model execution is pending. Replay uses an explicit simulation flag.



FEASIBILITY AND VIABILITY



Analysis of feasibility

A local PoC runs today. National MVP starts with a real NIST model pair.

Potential challenges and risks

A trigger test can raise a false alarm

Model files can execute unsafe code

Large models exceed browser uploads

Keys reset and replay checks are limited

Baseline assessment will not require retraining the supplied model

Strategies for overcoming challenges

Use clean controls and fixed thresholds

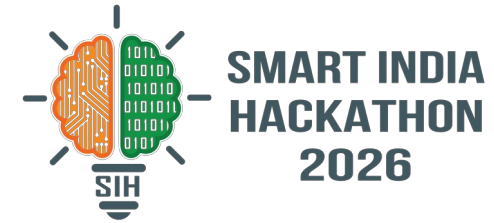
Load inside a restricted offline worker

Hash assets as streams and queue workers

Persist trusted keys and receiver state



IMPACT AND BENEFITS



Potential impact on the target audience

Defence AI reviewers and teams accepting externally supplied vision models

Operational benefit

Evidence helps analysts investigate unexpected model behaviour.

Governance benefit

Signed records expose later changes.
Offline use keeps inputs on site.

Economic benefit

Reusable checks may reduce repeated manual review and integration work.

How we will measure impact

Detection at a fixed false-alarm rate.
Review time, runtime and memory per model.

Expected benefits require validation. No field impact or cost saving claimed yet.



Research that grounds the problem and the validation plan

NIST TrojAI Round 13

128 training models in the round documentation. Four attack effects.
pages.nist.gov/trojai/docs/object-detection-feb2023.html

COCO-O, ICCV 2023

6,782 images across six domains for natural distribution shift.
github.com/alibaba/easyrobust/tree/main/benchmarks/coco_o

ObjectLab, 2023

Research on label errors in object-detection datasets.
arxiv.org/abs/2309.00832

Pichler et al., AISTATS 2024

Limits of universal backdoor detection motivate explicit coverage.
proceedings.mlr.press/v238/pichler24a.html